TRADEPULSE AI · INDEX OPTIONS · STRATEGY GUIDE

VWAP-EMA Scalper

A 1-minute scalping strategy for NIFTY, BANKNIFTY and SENSEX options. It reads the near-month future's own price and volume (a true VWAP) with the 5 and 9 EMAs, and confirms with option delta, open interest, PCR, the premium's volume profile and India VIX. The premium is picked from the OI data. The stop comes from structure through delta and gamma, and R:R is measured on the actual fill. Positions are sized on a stressed loss and managed on the server with a staged trailing stop.

Timeframe	Criteria	Risk : Reward	Execution
1-minute	**7 (10 pts, all needed)**	**1:1 or 1:2**	**Paper (Scalper)**

Status and scope

This document describes the strategy as implemented in the TradePlusAI repository (branch `claude/busy-planck-kqqyyn`). It has been tested on synthetic 1-minute data and has not yet run against live Groww market data. Automatic placement goes to the paper Smart Scalper only; live orders still need TradePulse's preview-and-PIN flow. This is software documentation, not investment advice.

Version 1.1 · 1 November 2026 (v1.1: true futures VWAP, fill-based R:R, stressed-risk sizing, explicit PCR zones, server-side exits) · Source: `quant/src/tradepulse_quant/algo_engine/scalp_strategy.py`, `services/paper-trading/src/smart-scalper.ts`, `apps/web/components/scalper/`

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1. The strategy at a glance

The VWAP-EMA Scalper buys an index option (a **CE** for a long, a **PE** for a short) when the 1-minute index tape and the option chain agree on a short, clean move. It enters only when **all seven criteria** pass, worth 10 points in total, and every V5 rule gate is green.

1. **Trend and location.** The near-month **future** is on the trade side of its own sloping session VWAP and is not stretched away from it.
2. **Momentum trigger.** The 5 EMA leads the 9 EMA, price pulls back to the 9 EMA, and a candle resumes the move on above-average futures volume.
3. **Option confirmation.** The selected strike has a delta of 0.40-0.60. Option writers (OI) and the PCR support the side. The premium itself is accepted above its point of control. India VIX is in a tradeable range.
4. **Plan.** The stop is placed on the chart (structure) and translated to the premium with **delta and gamma**. R is measured from the **actual fill** (ask plus slippage), and the target is fill + 1R or 2R. Lots are sized on a **stressed** loss (slippage on both legs plus a stop gap). A trade is refused if charges and slippage eat more than a quarter of the reward.
5. **Execution and management.** The trade is placed on the paper Smart Scalper, automatically or with one click, after a re-plan on the real fill. A **server-side monitor** then moves the stop to a cost-covered breakeven, locks profit and trails, with or without a browser open.

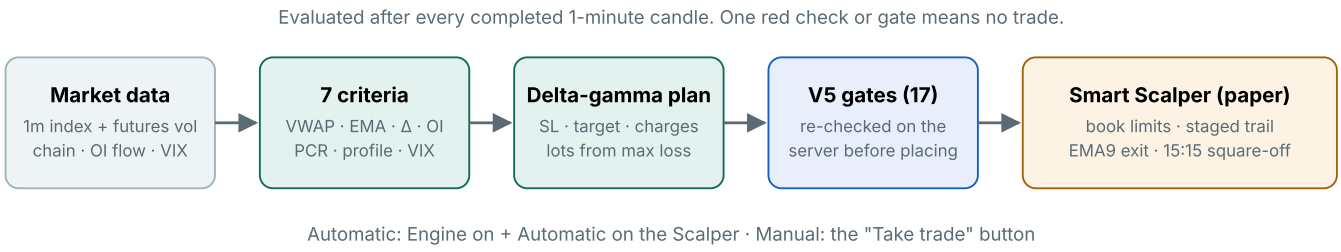


Figure 1. Decision pipeline. Each stage must pass before the next runs.

2. Data the strategy reads

All inputs are fetched server-side by `/api/algo-trading` and passed to the Python engine. If an input is missing, the criteria that need it **fail closed**: the strategy waits instead of guessing.

Input	Source in TradePulse	Used for
Near-month future, 1-minute OHLCV	Groww history (`volume=futures` returns `futuresCandles`)	The tape: **true VWAP** (the future's own price and volume), 5/9 EMA, ATR(14), trigger, structural stop. Without it, VWAP fails closed.
Index 1-minute candles and spot	Groww history and the chain's spot	Data-quality and gap gates; index levels = future levels − basis
Option chain (full, ±3%)	`/api/option-chain?full=1`	Premium, delta, IV, theta, OI, volume, lot and tick size for every strike
Options flow	Market-intel engine (`options_flow`)	Per-strike 5-minute OI change and build-up type, OI direction score, PCR and its 5-minute change, OI walls
India VIX	Market-intel volatility (value and regime)	Volatility criterion
Selected premium, 1-minute candles	`/api/market-data/option-history` (second pass)	The premium's volume profile (POC, VAH, VAL)
Operational state	Broker health, daily risk, contract master, SAFE_MODE, kill switch	V5 operational gates

> **Completed candles only**
> The engine drops any candle that is still forming before it evaluates. This applies to the index (1-minute buckets) and to the premium. A signal can therefore never repaint, and it never confirms on a half-built candle. The page re-runs the engine 4 seconds after every minute closes.

> **Two-pass evaluation**
> Pass 1 reads the tape and picks the strike from the OI data. If that strike's premium candles are not loaded yet, the engine returns `needs_premium_candles`. Pass 2 adds those candles so the volume-profile criterion can be scored.

3. Choosing the side

The side comes from where price sits relative to VWAP and how the two EMAs are stacked, on the latest completed candle:

Condition	Side	Buys
Close > VWAP **and** 5 EMA > 9 EMA	LONG	CE (call)
Close < VWAP **and** 5 EMA < 9 EMA	SHORT	PE (put)
VWAP and EMAs disagree	Lean only (the side of VWAP)	Nothing until both agree
Close exactly on VWAP	None	Nothing

Only option buying is used. The PE side is the exact mirror of the CE side throughout this document: above becomes below, rising becomes falling, highs become lows.

4. The seven criteria

Each criterion scores points. The strategy trades only at **10 / 10**, so every criterion is effectively mandatory. The score is shown so a trader can see how close a setup is. The rules below are for a long (CE); the short (PE) side mirrors them.

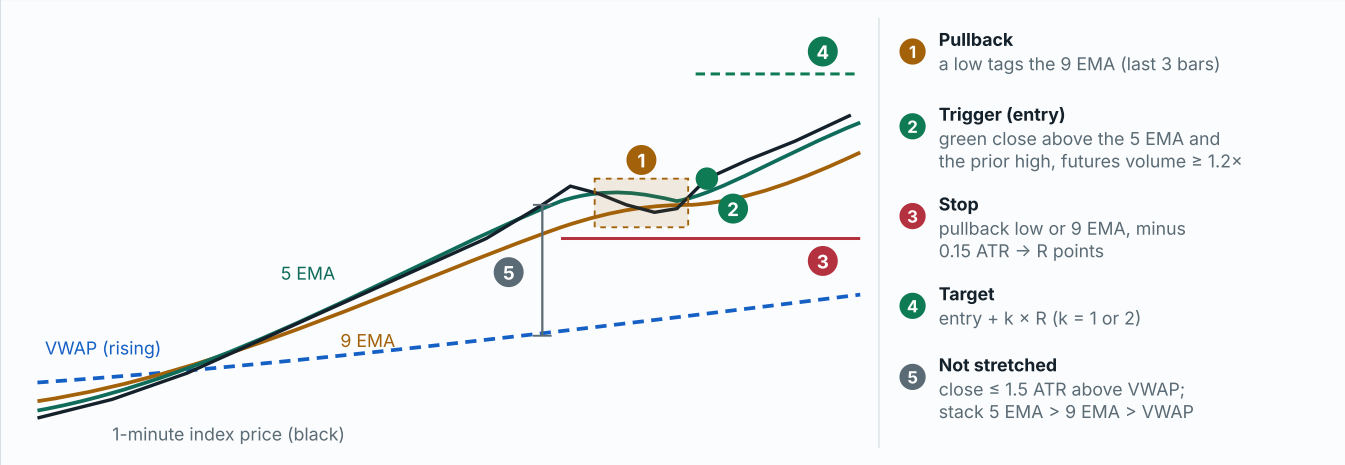


Figure 2. The long setup on the 1-minute index chart (schematic): trend above a rising VWAP, the EMA stack, the pullback to the 9 EMA, the resumption trigger, and the structural stop and target.

① VWAP

2 points

- **1 point:** the **future's** close is above its session VWAP **and** not more than **1.5 × ATR(1m)** away from it.
- **1 point:** VWAP has risen over the last **5 bars**.
- VWAP is computed on the **near-month future's own OHLCV**, the same instrument's price and volume. Weighting the index price by futures volume would only be a proxy. Without futures OHLCV, the criterion fails closed.

Why: VWAP is the session's fair price, where institutions benchmark fills. A long above a rising VWAP is with the session's flow. Stretching far above it is chasing, and those moves often snap back to VWAP inside a scalp's time frame.

② 5 / 9 EMA

2 points

- **1 point, the stack:** 5 EMA > 9 EMA > VWAP.
- **1 point, the trigger:** within the last 3 bars, a candle's low tagged the 9 EMA (tolerance 0.05 ATR). Then a candle that is **green**, closes **above the 5 EMA** and **above the previous candle's high**, on futures volume of at least **1.2×** the prior 20-bar average.
- **Freshness:** the trigger must be the latest completed candle, or the one before it if price hasn't run more than 0.5 ATR past the trigger close and is still above the 5 EMA.

Why: the 5/9 EMA pair is the classic fast momentum read on 1-minute charts. Buying the first resumption after a shallow pullback to the 9 EMA gives a defined stop just under the pullback. The volume check filters out drifts that nobody is trading.

③ Delta

1 point

- The auto-selected strike's **|delta| is between 0.40 and 0.60**, near the money.

Why: a scalp needs a premium that moves with the index point for point. Below 0.40 (OTM) the premium barely reacts and theta decay dominates. Above 0.60 (deep ITM) the option is a costly proxy for the future, with less leverage per rupee.

④ Open interest

2 points

- **1 point:** the 5-minute OI writer-flow score is **+1 or more**. Put writing or call short covering is outweighing call writing.
- **1 point:** at the ATM strike there is **supportive build-up**: put short build-up (put writing), call short covering, or call long build-up. There must also be **no fresh call writing** at the ATM, and the **largest call-OI wall** above spot must be farther away than the target (in index points).

Why: option writers are the best-capitalised participants. Put writers defend support; call writers cap rallies. A long needs writers leaning its way and no big call wall sitting between the entry and the target.

⑤ PCR (put-call ratio, OI)

1 point

- Explicit zones (inclusive bounds as written):

Zone	PCR	Long (CE)	Short (PE)
Stretched low	< 0.50	no	no
Bearish	0.50 ≤ PCR < 0.80	no	yes, if not rising
Neutral	0.80 ≤ PCR ≤ 1.20	yes, if not falling	yes, if not rising
Bullish	1.20 < PCR ≤ 1.60	yes, if not falling	no
Stretched high	> 1.60	no	no

- "Not falling" means a 5-minute PCR change ≥ 0, and "not rising" means ≤ 0. A missing change does not block. In the neutral zone the change decides the side.

Why: a healthy, rising PCR means puts are being written under the market (support). Extreme readings mean positioning is crowded and tends to unwind.

⑥ Volume profile of the selected premium

1 point

- Built from the **selected premium's own** 1-minute candles today: the point of control (POC) and the 70% value area (VAL to VAH).
- Passes when the premium closes **above its POC and** either at or above the **VAH**, or on a **volume burst of at least 1.5×** the prior 20 bars.

Why: the option must confirm the index. A premium accepted above the price where most of today's contracts traded shows buyers in control of that contract, not just the underlying.

⑦ India VIX

1 point

- VIX regime LOW, NORMAL or HIGH (not EXTREME) **and** a VIX value between **10 and 24**.

Why: below about 10, premiums hardly move and charges dominate. In extreme volatility, 1-minute structure breaks down, spreads widen and stops get run.

5. Auto-selecting the premium from OI

The strategy picks the contract itself; the trader doesn't choose a strike. Candidates are the trade side's options (calls for a long) with:

- |delta| between **0.35 and 0.65**. This is a wider net than the 0.40-0.60 the delta criterion requires, so near-misses can be shown.
- A premium of at least **₹20**. Where a bid/ask is available, a spread of at most **3%**; wider spreads are penalised.

Each candidate is ranked by a selection score:

```
score = 4 × (OI rank + volume rank) / (2 × (n - 1))   ← liquidity, 0-4
      + 2   if the option's 5-minute activity is LONG_BUILDUP or SHORT_COVERING
      - 2   if it is SHORT_BUILDUP (fresh writing of that option)
      + 2 × (1 - min(|delta - 0.5| / 0.15, 1))   ← closeness to delta 0.50, 0-2
      - 3   if the spread is wider than 3%
```

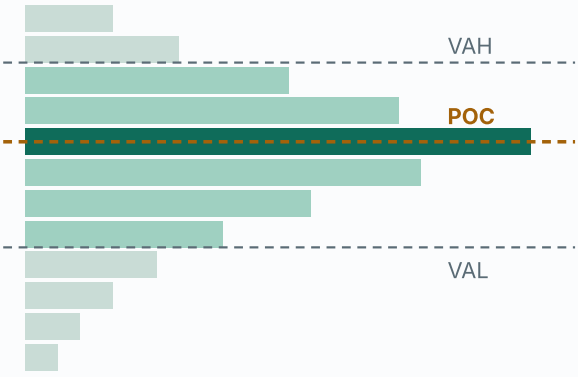
The top-ranked strike becomes the **selected premium**. On the Scalper desk the trade side's card and chart switch to it, marked with an `OI pick` badge. The ▲▼ buttons still override it until the next new pick. The next three candidates are listed with their scores.

6. The premium's volume profile

Each 1-minute premium candle's volume is spread evenly across the price bins its high-low range covers, using 24 bins over the day's range.

- **POC**: the bin with the most volume, where most contracts changed hands.
- **Value area**: grown outward from the POC, always adding the busier neighbouring bin, until it holds **70%** of the volume. Its top is the **VAH** and its bottom the **VAL**.

The criterion asks whether the premium is **accepted above value**: a close above the POC, plus either a break of the VAH or a 1.5× volume burst. The chart draws the POC, VAH and VAL as dashed lines.



volume at price → (70% value area shaded)

7. Stop, target and delta-gamma risk-reward

7.1 Structure first: the stop on the index

```
stop level   = min(pullback low, 9 EMA) - 0.15 × ATR(1m)        (long, on the future's
tape)
R (points)   = entry close - stop level
if R < 0.30 × ATR  → widen R to 0.30 × ATR     (tighter is inside 1-minute noise)
if R > 1.50 × ATR  → NO TRADE                   (too wide to be a scalp)
index levels = future levels - basis            (basis = future - index, near constant
intraday)
```

The stop is where the scalp is proven wrong: back through the pullback and the 9 EMA. Future points and index points move together intraday, so the distance carries over to the index, and from there to the premium.

7.2 Mapping index points to the premium with delta and gamma

An option premium does not move one-for-one with the index. For a small index move *x*, its change is approximately:

$$\Delta\text{premium}(x) \approx |\text{delta}| \cdot x + \tfrac{1}{2} \cdot \text{gamma} \cdot x^2$$

The stop comes from structure. R:R is measured on what the trader actually pays:

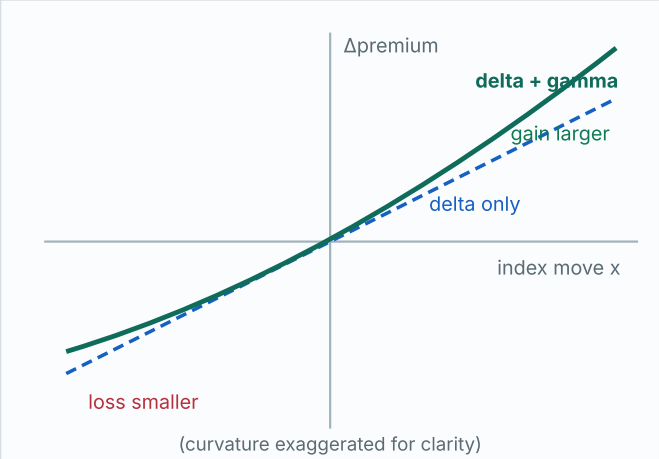
```
theta drag      = |theta per day| × 10 / 375                (375 trading minutes a
day)
premium SL      = M - ( |delta| · R - ½ · gamma · R² + theta drag )   ← structural,
from the market premium M
expected fill F = ask × (1 + slippage), rounded UP to the tick      ← what the scalper
pays
premium R       = F - premium SL
premium target  = F + k · premium R,   k = 1 (1:1) or 2 (1:2)       ← 1:1 / 1:2 hold on
the real fill
index move x needed for the target: |delta| · x + ½ · gamma · x² = target - M + theta
drag
delta from the chain;  gamma from Black-Scholes using the chain's IV and expiry
```

Because the fill pays the spread and slippage, the index has to move a little more than k × R to reach the target. The engine solves for that move *x* and checks it against the OI wall.

Why gamma matters. Gamma makes the premium's response convex. On an adverse move the long option loses a little *less* than delta alone predicts ($-\tfrac{1}{2}\gamma R^2$), which lifts the structural SL slightly. On a favourable move it gains a little *more*, so the index needs a little less distance to reach the premium target. Theta works against both.

Effect by expiry. With a week or more to expiry, gamma is small and the correction is a few paise. On expiry day gamma is large, so premium stops and targets move fast. The panel flags expiry days.

Both prices are rounded to the contract's tick (₹0.05).



7.3 The reward-to-risk gate

The V5 R:R gate compares the premium reward-to-risk **at the expected fill** with the ratio the trader chose, allowing 3% for tick rounding. Every other strategy in TradePulse keeps its 2.0 minimum.

Chosen	Premium target	R:R needed at the fill
1:2 (default)	fill + 2 × premium R	≥ 1.94
1:1	fill + 1 × premium R	≥ 0.97

8. Charges and position size

Every plan is costed per lot, for both exits (target and stop), with a model tuned for liquid near-ATM index options:

Component	Rate
Brokerage	₹20 per order (₹40 round trip)
STT	0.1% of the sell premium
Exchange transaction charge	0.03503% of premium turnover
SEBI fee / stamp duty	₹10 per crore / 0.003% of the buy premium
GST	18% on brokerage + exchange + SEBI
Slippage	The Smart Scalper's own fill slippage (`SCALPER_SLIPPAGE_PCT`, 0.3% per side by default), applied **adversely on both legs**: in the entry fill and on the exit
Spread	Entry: paid in the ask. Exit: half-spread at the bid. Capped at 0.5% of the premium per side (Groww's chain has no live bid/ask; the route estimates it from IV)

> **The 25% rule**
> If the fees, the exit spread and slippage, and what the fill paid above the market together exceed **25% of the reward**, the trade is refused: the move is too small to scalp profitably. A **1:1** NIFTY scalp with a 10-15 point stop is usually refused (30% in the worked example). **1:2** is the practical default; 1:1 passes when the structural move is larger, for example on BANKNIFTY (21% in the tests).

Planned and stressed loss, and lot sizing

```
planned loss / lot  = (F - SL) × lot + fees + exit half-spread + exit slippage
stop gap / unit     = max(2 ticks, 10% × premium R)
stressed loss / lot = planned loss / lot + stop gap × lot
lots                = ⌊ max loss per trade ÷ stressed loss / lot ⌋  (≤ lots setting)
```

Sizing on the **stressed** loss means a gap through the stop still fits the max loss. The defaults are a ₹1,500 max loss and 1 lot, from the Scalper desk's own controls. Both losses and both R:R figures (net and stressed) are shown before the trade.

9. Worked example (real engine output)

Inputs: a synthetic near-month NIFTY future 1-minute tape (morning chop, a measured trend, a pullback to the 9 EMA, then a high-volume resumption candle at 09:40), the index 60 points below it, and a synthetic chain and OI flow (the quant test fixture), run through the real engine. DEMO DATA, not live prices

Criterion	Reading	Points
VWAP	Future 24,879.00 above its VWAP 24,867.75 (index 24,819, basis 60), VWAP rising 2.4 pts in 5 min	2 / 2
5/9 EMA	5 EMA 24,876.23 > 9 EMA 24,875.21 > VWAP (future); pullback to the 9 EMA, resumed at 09:40 on 1.65× volume	2 / 2
Delta	NIFTY 24800 CE, delta 0.58 (picked from OI)	1 / 1
OI	Writer score +1; ATM 24800: call short covering, put short build-up; call wall 24950 is 131 pts away	2 / 2
PCR	1.05, +0.02 in 5 min: neutral zone, rising, so longs are allowed	1 / 1
Volume profile	Premium ₹127.80 above POC ₹127.26, on a 2.2× volume burst (VAH ₹129.10)	1 / 1
India VIX	13.2, normal	1 / 1
Total	**All seven met; all 17 V5 gates green**	**10 / 10**

Step	Calculation	Result
Index stop / R	structural stop 11.39 pts below (future 24,879.00 → index 24,819.00 − 11.39)	R = 11.39 pts
Greeks	delta 0.576 · gamma 0.000791 · theta ₹9/day · IV 13.5%	theta drag ₹0.24
Premium SL (structure)	market ₹120.45 − (0.576 × 11.39 = ₹6.56 − gamma ₹0.05 + theta ₹0.24)	₹113.70
Expected fill	ask ₹120.85 + 0.3% slippage, rounded up to the tick	₹121.25
Premium R / target (1:2)	R = 121.25 − 113.70 = ₹7.55; target = 121.25 + 2 × 7.55	₹136.35 (2.00R on the fill)
Index move needed	$0.576 \cdot x + \frac{1}{2} \cdot 0.000791 \cdot x^2 = ₹16.14$ → x (vs 22.79 structural; OI wall 131 pts away ✓)	27.5 pts
Costs per lot	fees + exit half-spread ₹89.24, exit slippage ₹26.59; entry slippage and spread are in the fill	16% of the reward ✓
Planned / stressed loss	7.55 × 65 + ₹109.33 = ₹600.08; + stop gap ₹0.75 × 65	₹600 / ₹649
Net / stressed R:R	(15.10 × 65 − ₹115.83) ÷ planned or stressed loss	1.44 / 1.33
Size	⌊₹1,500 ÷ ₹649⌋ = 2, capped at the 1-lot setting	1 lot = 65 qty
Fill tolerance	max(2 ticks, 0.2 × ₹7.55) above the expected fill; worst-fill target capped at ₹140.85	max entry ₹122.75

Same tape at 1:1: target ₹128.80, charges and slippage 30% of the reward. The trade is **refused** by the 25% rule.

10. The V5 rule gates

A confirmed strategy decision must also pass TradePulse's authoritative no-trade engine. One red gate blocks the trade. For this strategy the generic gates map as follows:

Gate	For the VWAP-EMA scalper
Data quality	Valid, ordered candles; the latest under 15 minutes old during market hours
Session	Inside the entry window (09:35-14:45 IST; earlier cut-off on expiry day)
Option quote, daily risk, broker, contract metadata, reconciliation, execution	Operational state, supplied and verified on the server
Strategy	All seven criteria met and a valid plan
"Breakout" → **VWAP**	Criterion ① passed
"Retest" → **5/9 EMA trigger**	Criterion ② passed
VIX	Regime LOW / NORMAL / HIGH
Gap	Normal open, or a confirmed gap hold/fill
Score	10 / 10
Liquidity	The selected strike scores ≥ 2/3 (premium ≥ floor, OI > 0, traded today)
R:R	Premium R:R ≥ 0.97 (1:1) or ≥ 1.94 (1:2)
Charges vs reward	Charges ≤ 25% of the reward, and lots ≥ 1
SAFE_MODE / kill switch	Either one blocks all new entries

Not applied, by design: the 5-minute ADX regime gate (the 1-minute VWAP/EMA stack is the scalper's own trend filter) and the generic option-flow gate (OI and PCR are strategy criteria ④ and ⑤ instead).

11. Placing the trade on the Scalper

Scalper setting	Behaviour
Strategy: **VWAP-EMA scalper** (default)	The Scalper trades this strategy. The level engine never enters, but it still manages exits.
Engine **on** + Trade **Automatic**	A confirmed setup is placed automatically, once per setup (keyed on the trigger candle and side).
Engine **on** + Trade **Manual**	"⚡ Take trade · BUY 24800 CE 1:2" lights up when confirmed; one click places it.
Strategy: Level scalper	The desk's original support/resistance engine.

Every placement takes the same safe path:

1. **Server re-check.** `POST /api/algo-trading` with `action: PLACE_ON_SCALPER` re-runs the strategy and all V5 gates on the server. The browser can't confirm a trade.
2. **Book limits.** The Smart Scalper applies its own limits: kill switch, SAFE_MODE, daily loss limit, a 5-minute cool-down after a loss, a stop after 3 consecutive losses, the trade cap, one engine position per index, and 09:15-15:15 IST on trading days only.
3. **Re-plan on the actual fill.** The fill is the live ask plus slippage, rounded against the trader.
   - A fill above the plan's **max entry** (expected fill + tolerance) is refused as a stale setup.
   - The **stop stays at its structural price**; it is never moved by slippage.

- R and the target are re-derived from the fill (target = fill + k × R), but never beyond the OI-wall cap, and the R:R must still reach the minimum.
- Lots are re-sized so the **stressed** loss at that fill fits the max loss.
- The position records the planned versus actual fill and the planned and stressed loss.

> **Paper only**
> Automatic placement never sends a live order. Real-money orders remain on TradePulse's separate live path: `/api/live-orders`, with a preview, a PIN, the LIVE limits and the kill switch.

12. Trade management: SL, target and trailing SL

Once in, the position is managed by a **staged, cost-covered trail**. R is the initial premium risk (entry − initial SL). The stop only ever moves **up**.

Stage	Triggered when the best premium reaches	SL moves to	Example (fill ₹121.25, R ₹7.55)
Initial	entry	the structural (delta-gamma) SL	₹113.70
Breakeven	+0.5R (₹125.03)	entry + round-trip costs per unit (cost-covered)	₹123.03
Locked	+1R (₹128.80)	entry + 0.5R	≈ ₹125.03
Trailing	beyond +1R	best premium − 0.6R	at ₹132 → ₹127.47

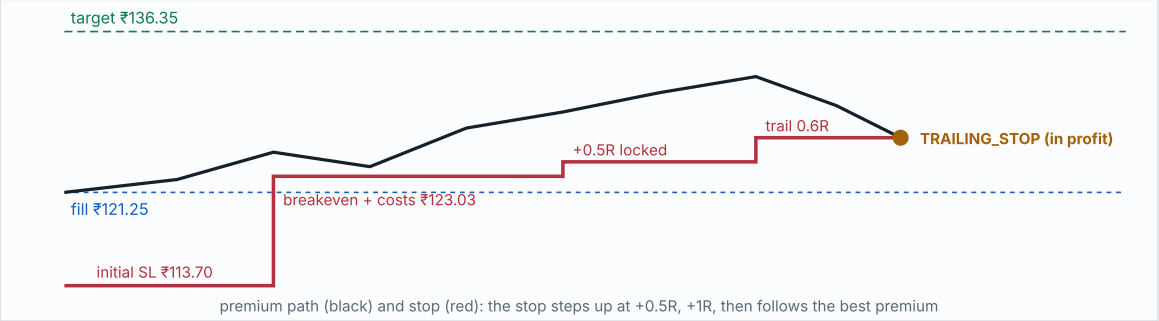


Figure 3. The staged trail on a winning scalp.

Exits, in priority order

Exit reason	Rule
`SQUARE_OFF_15_15`	Every open position is closed at 15:15 IST
`MAX_LOSS_PER_TRADE`	The open loss reaches the max loss per trade
`SPOT_LEVEL_BROKEN` / `SPOT_TARGET`	The index trades through the structural stop / reaches the index target
`TARGET`	The premium reaches the premium target
`STOP_LOSS`, `BREAKEVEN_STOP`, `LOCKED_PROFIT_STOP`, `TRAILING_STOP`	The premium reaches the stop; the reason names the trail stage
`EMA9_LOST`	After breakeven: a **completed** 1-minute index candle closes back below the 9 EMA (above it, for a PE). The scalp's reason is gone.
`SCALP_TIME_STOP`	The trade hasn't reached +0.5R within the Scalper's time stop (default 12 minutes)

The trail stage, the initial SL and the costs per unit are saved with the order, so a server restart resumes the same trail. On the premium chart the risk-reward drawing follows the live SL. Once the SL is above the entry, it shows the profit locked, e.g. "SL · +0.5R locked ₹124.32 (+₹225.55)".

> **Exits run on the server**
> A server-side monitor (`apps/web/lib/scalper-monitor.ts`) runs in the Node server process every 5 seconds. It fetches the live option quotes, the index spot and 1-minute candles itself, and applies every exit above. It starts at server boot (`instrumentation.ts`), which also reloads persisted open positions after a restart, so a closed, sleeping or disconnected browser no longer leaves a position unmanaged. A missing spot quote never triggers an index exit, and the 15:15 square-off runs even without a fresh quote. The desk shows the monitor's status, and its own 10-second scan still runs; the scalper serialises both, so a position is never closed twice.

13. Screens

All screens below show DEMO DATA, not live market prices. They were captured with v1.0 of the plan (ask-based entry), so their premium figures differ slightly from section 9. The layout is unchanged in v1.1, which adds planned/stressed loss, fill tolerance and the server exit-monitor status.

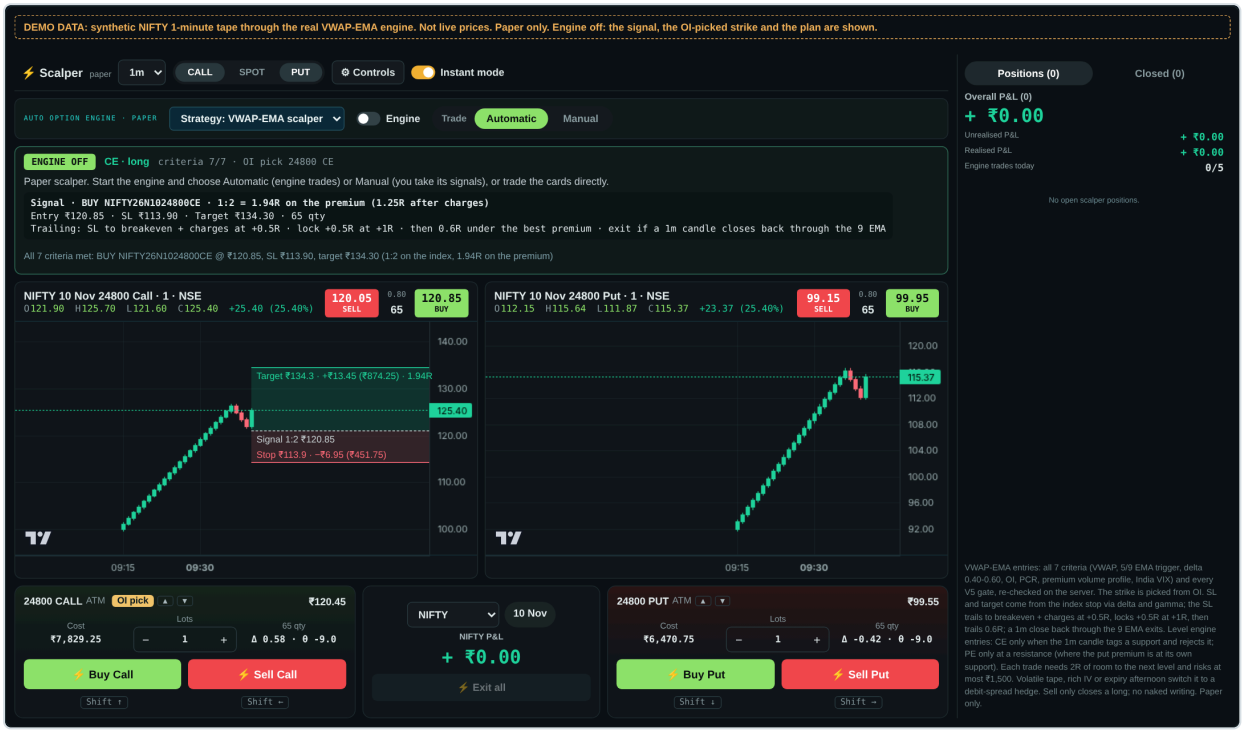


Figure 4. The Scalper desk with Strategy set to VWAP-EMA: criteria 7/7, the OI-picked 24800 CE (badge), the plan (entry ₹120.85, SL ₹113.90, target ₹134.30) and the risk-reward drawing on the call chart.

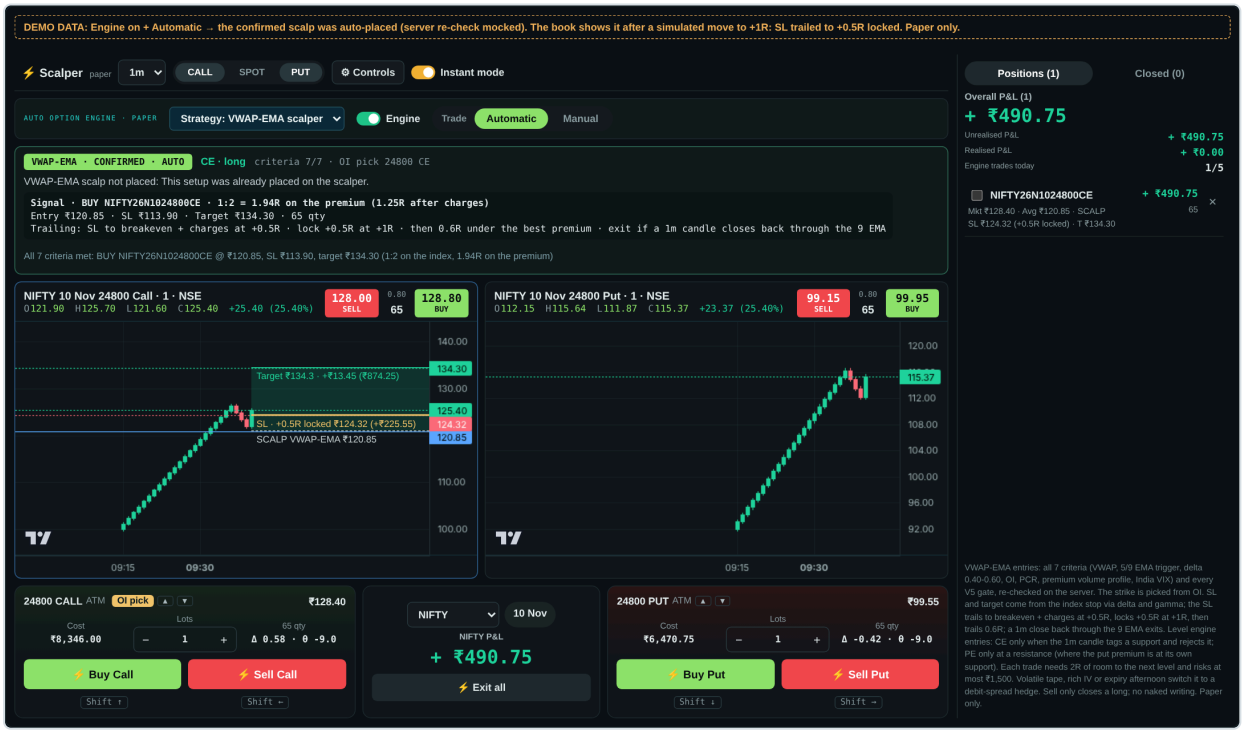


Figure 5. Engine on + Automatic: the scalp was placed (server re-check mocked for the demo). Shown after a simulated move to +1R: the SL has trailed to "+0.5R locked" and the position cannot lose. A second attempt on the same setup is refused as already placed.



Figure 6. The strategy panel on the execution page: the 1:1 / 1:2 selector, criteria tiles and reasons, the OI-selected premium with its Greeks, the delta-gamma working of the SL and target, the premium chart with POC / VAH / VAL and the risk-reward zones, and the 17 V5 gates.

14. Parameter reference

Defaults in `ScalpConfig` (Python) and `VWAP_EMA_TRAIL` (TypeScript). The max loss per trade and the lots cap come from the Scalper desk's Controls.

Parameter	Value	Meaning
EMA fast / slow	5 / 9	Momentum pair on 1-minute closes
ATR period	14	1-minute ATR used for every distance rule
Minimum session bars	15	No evaluation before 15 one-minute candles today
VWAP slope bars	5	VWAP must rise (fall) over this many bars
Max distance from VWAP	1.5 ATR	Beyond this the entry is a chase
Pullback window / EMA tolerance	3 bars / 0.05 ATR	Where the 9 EMA tag must occur
Trigger volume	≥ 1.2×	Futures volume vs the prior 20 bars
Trigger age / max drift	≤ 1 bar / 0.5 ATR	Freshness of the signal
Strike pool delta / criterion delta	0.35-0.65 / 0.40-0.60	Selection band / pass band
Min premium / max spread	₹20 / 3%	Candidate filters
PCR zones	0.50 / 0.80 / 1.20 / 1.60	Stretched < 0.50 ≤ bearish < 0.80 ≤ neutral ≤ 1.20 < bullish ≤ 1.60 < stretched
India VIX	10-24, not EXTREME	Tradeable volatility
Profile bins / value area / burst	24 / 70% / 1.5×	Premium volume profile
Stop buffer / min R / max R	0.15 / 0.30 / 1.50 ATR	Structural stop rules
Hold for theta	10 min	Theta charged on the plan
Max charges share	25%	Of the reward at the target
Slippage (both legs) / spread cap	0.3% / 0.5% per side	Same as the Smart Scalper's fills (`SCALPER_SLIPPAGE_PCT`)
Stop gap (stressed risk)	max(2 ticks, 10% R)	Added to the planned loss for sizing
Fill tolerance	max(2 ticks, 20% R)	Max fill above the expected fill
R:R tolerance	3%	Tick-rounding allowance on the premium ratio at the fill
Exit monitor interval	5 s	Server-side exit checks
Trail: breakeven / lock / lock to / gap	+0.5R / +1R / +0.5R / 0.6R	Staged trailing stop
Max loss per trade / lots cap	₹1,500 / 1	Scalper Controls (defaults)
Scalp time stop	12 min	Scalper Controls (default)

15. Limitations and first-live checklist

- **Not yet run on live data.** The build environment could not reach market data. The strategy is covered by tests on synthetic tapes (29 Python and 23 TypeScript tests for the strategy, fill re-planning, stressed sizing,

PCR zones, futures VWAP, the trail, the server monitor and the chart), and the full suites pass. Watch the first live sessions in paper mode.

- **Check on the first live sessions:**
  - the near-month future's 1-minute OHLCV arrives (`futuresCandles`), so VWAP is observable;
  - market-intel's per-strike 5-minute OI change and build-up are populated (they need a 5-minute baseline after the open);
  - the selected premium's 1-minute candles load for the volume profile;
  - the chain's delta and IV look sane on expiry day.
- **Spread estimate.** Groww's chain carries no live bid/ask, so the spread is estimated from IV and capped. Real fills may differ, which is why placement re-plans on the actual fill.
- **Futures data dependency.** Without the near-month future's 1-minute OHLCV the VWAP criterion fails closed (no trades), by design.
- **1:1 rarely qualifies on NIFTY** because of the 25% charges rule, now that slippage is counted on both legs. That is by design, not a fault.
- **Server-side exits are paper only.** The monitor manages the paper Smart Scalper book. Live orders use TradePulse's separate live monitor with broker stop orders.
- **Paper first.** Run the strategy in paper mode for several weeks and review the closed-trade log (exit reasons, R multiples, charges) before considering live trading through TradePulse's PIN-confirmed flow.

> **Risk disclaimer**
> Options trading involves substantial risk, and most intraday traders lose money. This document describes software behaviour; it is not investment advice, and TradePulse is not a SEBI-registered adviser. Live API trading may require static-IP registration and algo approval with your broker under SEBI's retail algo framework.

Revision history

Version	Change
1.1	Fixes from an independent review: • R:R measured on the actual fill, with a structural stop and a fill tolerance; • placement re-plans on the real fill; • slippage counted on both legs and lots sized on the stressed loss; • explicit PCR zones; • true VWAP on the near-month future's OHLCV; • exits moved to a server-side monitor.
1.0	First release.

Appendix: code and tests

File	Role
`quant/src/tradepulse_quant/algo_engine/scalp_strategy.py`	Criteria, strike selection, volume profile, delta-gamma plan, charges, sizing
`quant/src/tradepulse_quant/algo_engine/engine.py`	Wiring into the V5 engine (`_scalp_evidence`); completed-candle filtering
`quant/src/tradepulse_quant/algo_engine/no_trade_engine.py`	The V5 gates; strategy-specific R:R minimum

apps/web/app/api/algo-trading/route.ts	Inputs, two-pass evaluation, PLACE_ON_SCALPER with the server re-check
services/paper-trading/src/smart-scalper.ts	place() with the fill re-plan, manage(), vwapEmaTrail(), emaMomentumLost(), exits
apps/web/lib/scalper-monitor.ts, apps/web/instrumentation.ts	Server-side exit monitor, started at boot
apps/web/components/scalper/scalper-desk.tsx	Strategy selector, auto and manual placement, OI-pick strike, drawings
apps/web/components/scalper/premium-chart.tsx	Risk-reward drawing (reward zone, risk zone, locked-profit line)
quant/tests/test_scalp_strategy.py	Criteria, delta-gamma maths, 1:1 vs 1:2, PE mirror, V5 integration
tests/unit/scalp-place.test.ts, tests/unit/scalper-monitor.test.ts, tests/contract/algo-trading-route.test.ts	Placement limits, fill re-plan, stressed sizing, staged trail, momentum exit, server monitor, chart layout, route guards
docs/operations/vwap-ema-scalper.md	Operator notes (Markdown)